

Medical Named Entity Recognition (MedNER): A deep learning model for recognizing medical entities (drug, disease) from scientific texts

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Abstract—Medical Named Entity Recognition (MedNER) is an indispensable task in biomedical text mining. NER aims to recognize and categorize named entities in scientific literature, such as genes, proteins, diseases, and medications. This work is difficult due to the complexity of scientific language and the abundance of available material in the biomedical sector. Using domain-specific embedding and Bi-LSTM, we propose a novel NER model that employs deep learning approaches to improve the performance of NER on scientific publications. Our model gets 98% F1-score on a curated data-set of Covid-related scientific publications published in multiple web of science and pubmed indexed journals, significantly outperforming previous approaches deployed on the same data-set. Our findings illustrate the efficacy of our approach in reliably recognizing and classifying named entities (drug and disease) in scientific literature, opening the way for future developments in biomedical text mining.

Index Terms—NER, NLP, MedNER, medical named entity recognition, named entity recognition, Bi-LSTM, Deep learning in healthcare, Medical Natural Language Processing

I. INTRODUCTION

Medical Named Entity Recognition (MedNER) is a sub-field of Natural Language Processing (NLP) that focuses on identifying and categorizing entities within a medical text. The field of medical named entity recognition (NER) has garnered significant attention in recent years, as the ability to accurately identify and classify medical entities such as diseases, symptoms, and treatments within unstructured medical text has numerous applications in the healthcare domain [1], [2]. This task is crucial for various applications in the medical domain, such as information extraction, medical knowledge base construction, and clinical decision support.

The objective of medical NER in drug and disease recognition is to precisely identify and classify these items within the scientific literature in order to facilitate the study and

interpretation of the text. This can be utilized for a variety of purposes, including the identification of potential drug-drug interactions, the creation of new medicines, and the analysis of disease prevalence and treatment trends.

Medical named entities are specific words or phrases that refer to distinct medical concepts or objects. These entities can include diseases, drugs, anatomical structures, and other medical terms. Recognizing these entities from scientific text is an important task in the field of medical natural language processing. By identifying and classifying medical named entities, we can extract valuable information from scientific literature and aid in decision making and research. This task is not always straightforward, as medical terminology can be complex and varied, but advances in machine learning and natural language processing have made it possible to accurately identify and categorize medical named entities.

Recognizing medical named entities in scientific text is crucial for multiple reasons. First, it facilitates the accurate and rapid extraction of important information from scientific literature. This is useful for a variety of purposes, including medical research, diagnosis, and treatment. Second, it identifies and organizes various medical entities, such as diseases, symptoms, treatments, and medications, so that individuals can better understand and treat medical conditions. Thirdly, it improves the accuracy and precision of natural language processing algorithms, which may be utilized for a variety of medical activities, including summarizing text, answering queries, and extracting information.

Overall, recognizing medical named entities from scientific text can play a crucial role in advancing the field of medicine and improving the quality of healthcare services. In this study, we propose a novel model for medical NER using deep learning techniques. Our model incorporates domain-

specific word embedding and bidirectional long short-term memory (Bi-LSTM) [3] networks to effectively capture contextual information and improve the performance of NER on scientific texts. Our proposed method outperforms existing state-of-the-art approaches in terms of precision, recall, and F1, demonstrating its effectiveness in accurately identifying and categorizing medical entities such as drugs and diseases within scientific texts.

II. RELATED WORKS

One recent scientific work on medical named entity recognition using Bi-LSTM is “Named Entity Recognition From Biomedical Texts Using a Fusion Attention-Based BiLSTM-CRF” by H. Wei et al. [4]. In this work, the authors propose an attention based named entity recognition system that utilizes Bi-LSTM-CRF networks to accurately identify medical entities in clinical notes. The system is able to extract different named entities from texts, such as DNA, RNA, Cell Type, Cell Line, and Protein, and achieves an F1-score of 73.5% for the NER task.

Another recent scientific work on this topic is “Combining contextualized embeddings and prior knowledge for clinical named entity recognition: evaluation study” by Jiang et al. [5]. In this paper, the authors propose a hybrid method for recognizing named entities in medical text. This method combines contextualized embeddings like ELMo and Flair to get better results. The authors found that the combination of these methods outperformed either method alone, and showed promising results on a variety of medical named entity recognition tasks, achieving a F1-score of 87.44%.

Another scientific work on medical named entity recognition using Bi-LSTM is the study by Xu, Kai, et al. [6] titled “A bidirectional LSTM and conditional random fields approach to medical named entity recognition”. In this study, the authors proposed a new model that combines a Bi-LSTM and a CRF (Conditional Random Field) for improved performance on medical named entity recognition tasks along with character embedding. The model was evaluated on several medical data-sets and showed improved performance compared to traditional models with an F1-score of 80.22%.

Another recent scientific work on medical named entity recognition using Bi-LSTM is the study by Chen, Xianglong, et al. [7] titled “Improving the named entity recognition of Chinese electronic medical records by combining domain dictionary and rules”. In this study, the authors proposed an improved Bi-LSTM-CRF model that incorporated additional features, such as word embedding and dictionary feature embeddings. The improved model showed better performance on medical named entity recognition tasks in Chinese electronic medical records compared to previous models with an F1-score of 90.65%.

Furthermore, a study by Li, Luqi, et al. [8] titled “An attention-based deep learning model for clinical named entity recognition of Chinese electronic medical records” proposed an attention-based Bi-LSTM-CRF model named BiLSTM-Att-CRF for medical named entity recognition from Chinese

electronic medical records. The model used an attention mechanism to focus on the most relevant contextual information for named entity recognition, leading to improved performance on medical named entity recognition tasks and achieved 90.29% F1-score.

Overall, these studies show the potential of using Bi-LSTM models for medical named entity recognition and highlight the effectiveness of combining Bi-LSTM with other techniques such as domain specific embeddings for improved performance. These studies also left open the idea that deep learning techniques could be used to improve the performance of named entity recognition in the medical domain.

III. METHODOLOGY

To develop the proposed deep neural network model several phases of operations are executed. In the first phase, some relevant scientific articles in pdf format are collected and converted to plain text format while maintaining proper sentence boundaries [9]. A data-set is then vetted by domain experts through annotation and validation. The data-set is generated in IOB format. After the data-set generation, the MedNER model is developed, trained, tested and tuned to ensure that it is able to accurately recognize named entities in unseen text. In the last step, the proposed model is used to pull named entities out of scientific texts. Fig. 1 depicts an overview of the methodology employed in this study.

A. Data acquisition and processing

Scientific articles are downloaded through the electronic resource of Universiti Malaysia Pahang’s library [10]. Using the search term “Drugs and diseases AND COVID-19” top 500 top cited scientific articles are accumulated and processed to be used in this study.

B. Data-set generation

To prepare the data-set domain expert used 50 relevant articles. From these articles, 3145 sentences are annotated with 5 labels, namely, “B-drug”, “I-drug”, “B-disease”, “I-disease”, and “O”. After the annotation process, the output is saved in IOB format. In the annotated data-set there are 4555 tokens with disease class, which is the combination of B-disease and I-disease labels, and 1205 tokens with drug class, which is the combination of B-drug and I-drug labels. The inter-annotator agreement (IAA) score for this data-set is 96% of Cohen’s kappa, which shows a high level of agreement between the annotators while annotating different labels in the texts. The data-set is available at Mendeley Data [11].

C. MedNER: deep neural network model

The proposed model consists of five layers, including three hidden layers. Among the five layers, the first one is the input layer, the second one is a domain-specific integer-coded word embedding layer, the third one is a dropout layer, the fourth one is a Bi-LSTM layer with TanH activation and 200 neurons, and the last one is a time-distributed dense layer with a softmax activation. The architecture and layer-wise

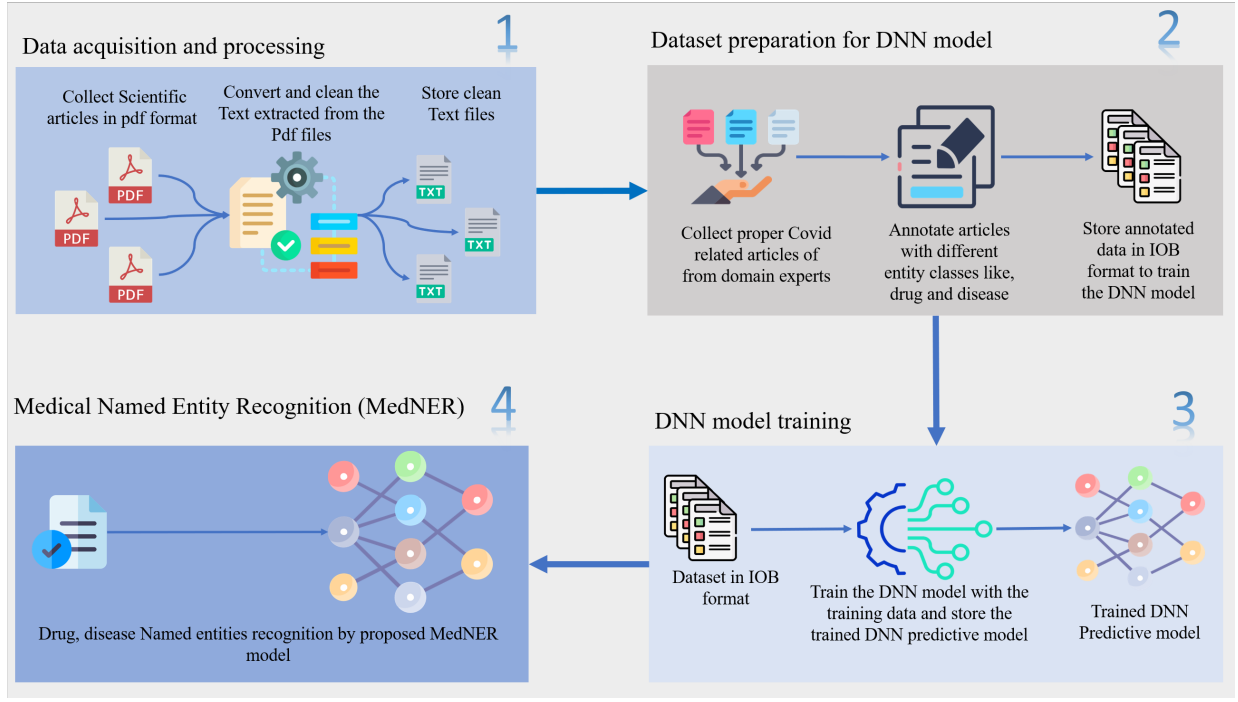


Fig. 1. Different Phases of the Methodology Employed in this Study.

configuration are shown in Algorithm 1. An overview of the architecture is shown in Fig. 2.

Algorithm 1 Proposed MedNER model architecture

- 1: Input: Sequence with token and labels: SL
- 2: num_words = unique words in data-set, num_tags = unique labels in data-set
- 3: set $maxLen = 60$
- 4: For $\{S \text{ in } sent_list\}$ { X = pad sequences words }
- 5: For $\{S \text{ in } sent_list\}$ { y = pad sequences labels, y = hot encode y }
- 6: Split data into training and testing set
- 7: $input_word = (maxLen)$
- 8: $Embedding(input_dim = num_words, output_dim = maxLen, input_length = maxLen)$
- 9: $SpatialDropout1D(0.2)$
- 10: $Bidirectional(LSTM(units = 200, return_sequences = True, recurrent_dropout = 0.2))$
- 11: $out = TimeDistributed(Dense(activation = 'softmax'))$
- 12: $matrec = Model(input_word, out)$
- 13: $matrec.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = [accuracy, precision_mt, recall_mt, f1_mt])$
- 14: $matrec.fit(train_test \text{ data}, batch_size = 16, verbose = 1, epochs = 50,$
- 15: $validation_split = 0.2, callbacks=[tensorboard_cbk, es])$
- 16: $model.save('medner.h5')$
- 17: Return $medner.h5$

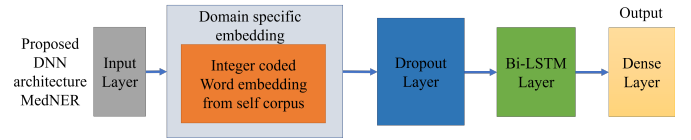


Fig. 2. Architecture of the proposed MedNER model

D. Evaluation metrics

Precision(P_r), Recall(R_c), and $F1$ are used to evaluate the proposed MedNER model. If each token is successfully detected during the analysis of entity predictions, the entity is marked as true. True Positives ($TPos$) are computed when the entities are correctly predicted; False Positives ($FPos$) are indicated when the predicted beginning token does not match the marked token of the entity. False Negatives ($FNeg$) are recorded when the system predicts the first token of the expected entities incorrectly. The equations stated in equation (1), (2), and (3) are used to determine the mentioned evaluation metrics.

$$P_r = \frac{TPos}{TPos + FPos} \quad (1)$$

$$R_c = \frac{TPos}{TPos + FNeg} \quad (2)$$

$$F1 = \frac{2 * P_r * R_c}{P_r + R_c} \quad (3)$$

IV. RESULT AND DISCUSSION

This study discusses the effectiveness of using machine learning algorithms for medical named entity recognition. The study compares the performance of several algorithms, including BERT [12], sciBERT [13], DecisionTree [14], RandomForest [15], and Gaussian NaiveBayes [16], on a dataset of published scientific articles. The experimental result is shown in Table I.

TABLE I
PRECISION, RECALL AND F1 SCORES ACHIEVED FROM THE EXPERIMENT
FOR DIFFERENT MODELS

Model	Precision	Recall	F1
Proposed MedNER	0.9843	0.9817	0.98
BERT	0.79	0.85	0.82
sciBERT	0.9	0.84	0.87
DecisionTree	.91	.98	.94
RandomForest	.92	0.96	.94
GaussianNaiveBayes	.90	.91	.904

The results show that proposed MednER model based on Bi-LSTM performed the best, achieving an F1 score of 0.98, followed by DecisionTree and RandomForest with an F1 score of 0.94 and sciBERT with an F1 score of 0.87. The study also found that the performance of the algorithms improved when the training data-set was domain specific and large BERT and sciBERT models performed less competently although they were fine tuned with the same data-set. This observation creates an opportunity to investigate large language models further with an increasing number of domain-specific data whether the performance can be improved or not. On the other hand, the experimental result shows that, domain-specific embedded neural network model without pre-trained embedding and language models performs better than the fine tuned large pre-trained models for drug and disease named entities recognition for covid-19 relevant scientific papers. Fig. 3(a), Fig. 3(b), Fig. 3(c), and Fig. 3(d) present the training and validation curves of the proposed MedNER model for loss, precision, recall, and F1 measures respectively. From the training and validation curves, it can be seen that, the proposed model does not suffer from over-fitting or under-fitting issue and has good convergence.

From the experimental results it can be concluded that, proposed model is the most effective algorithm among the compared ones for medical NER from scientific text. However, a note need to be taken that further research is needed to evaluate the performance of the proposed model on different types of medical text like electronic medical records and clinical decision support systems.

V. CONCLUSION AND FUTURE WORK

In conclusion, the study demonstrated the effectiveness of the proposed deep learning-based approach for medical named entity recognition. The experimental results showed that the proposed approach outperformed traditional machine learning methods and achieved a high level of accuracy in recognizing

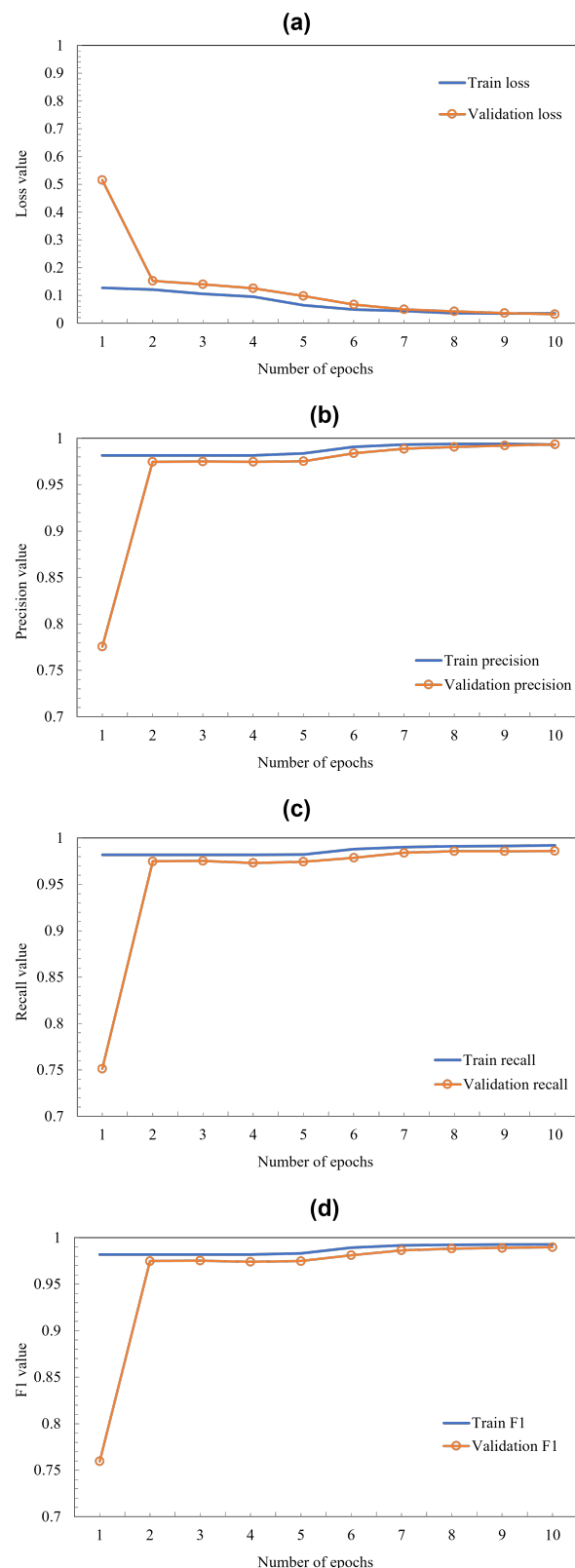


Fig. 3. Learning curves for training and validation phases of the proposed model. (a) shows the loss curves, (b) shows the precision curves, (c) shows the recall curves, and (d) shows the F1 curves.

different types of medical named entities such as drugs and diseases.

As future work, the plan is to extend the study to other medical domains and languages to further validate the proposed approach. Additionally, the proposed model can integrate the named entity recognition system with other natural language processing techniques to improve the overall performance of the system. Furthermore, the authors will explore the use of the proposed approach in real-world medical applications, such as electronic medical records and clinical decision support systems.

REFERENCES

- [1] P. Bose, S. Srinivasan, W. C. Sleeman IV, J. Palta, R. Kapoor, and P. Ghosh, "A survey on recent named entity recognition and relationship extraction techniques on clinical texts," *Applied Sciences*, vol. 11, no. 18, p. 8319, 2021.
- [2] B. Song, F. Li, Y. Liu, and X. Zeng, "Deep learning methods for biomedical named entity recognition: a survey and qualitative comparison," *Briefings in Bioinformatics*, vol. 22, no. 6, p. bbab282, 2021.
- [3] A. Graves and J. Schmidhuber, "Framewise phoneme classification with bidirectional lstm and other neural network architectures," *Neural networks*, vol. 18, no. 5-6, pp. 602–610, 2005.
- [4] H. Wei, M. Gao, A. Zhou, F. Chen, W. Qu, C. Wang, and M. Lu, "Named entity recognition from biomedical texts using a fusion attention-based bilstm-crf," *IEEE Access*, vol. 7, pp. 73 627–73 636, 2019.
- [5] M. Jiang, T. Sanger, X. Liu *et al.*, "Combining contextualized embeddings and prior knowledge for clinical named entity recognition: evaluation study," *JMIR medical informatics*, vol. 7, no. 4, p. e14850, 2019.
- [6] K. Xu, Z. Zhou, T. Hao, and W. Liu, "A bidirectional lstm and conditional random fields approach to medical named entity recognition," in *International Conference on Advanced Intelligent Systems and Informatics*. Springer, 2017, pp. 355–365.
- [7] X. Chen, C. Ouyang, Y. Liu, and Y. Bu, "Improving the named entity recognition of chinese electronic medical records by combining domain dictionary and rules," *International Journal of Environmental Research and Public Health*, vol. 17, no. 8, p. 2687, 2020.
- [8] L. Li, J. Zhao, L. Hou, Y. Zhai, J. Shi, and F. Cui, "An attention-based deep learning model for clinical named entity recognition of chinese electronic medical records," *BMC Medical Informatics and Decision Making*, vol. 19, no. 5, pp. 1–11, 2019.
- [9] M. S. U. Miah, J. Sulaiman, T. B. Sarwar, A. Naseer, F. Ashraf, K. Z. Zamli, and R. Jose, "Sentence boundary extraction from scientific literature of electric double layer capacitor domain: tools and techniques," *Applied Sciences*, vol. 12, no. 3, p. 1352, 2022.
- [10] UMP Library, "e-Resource UMP Lib," <https://login.ezproxy.ump.edu.my/login>, 2022. [Online]. Available: <https://login.ezproxy.ump.edu.my/login>
- [11] M. S. U. Miah, J. Sulaiman, and T. B. Sarwar, "Matrec: Material and process named entity recognition dataset for edlc," <https://data.mendeley.com/datasets/46p8vp9xts>, 2022.
- [12] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186. [Online]. Available: <https://aclanthology.org/N19-1423>
- [13] I. Beltagy, K. Lo, and A. Cohan, "Scibert: A pretrained language model for scientific text," *arXiv preprint arXiv:1903.10676*, 2019.
- [14] L. Rokach, "Decision forest: Twenty years of research," *Information Fusion*, vol. 27, pp. 111–125, 2016.
- [15] T. K. Ho, "Random decision forests," vol. 1, 1995, pp. 278–282.
- [16] I. Rish *et al.*, "An empirical study of the naive bayes classifier," in *IJCAI 2001 workshop on empirical methods in artificial intelligence*, vol. 3, no. 22, 2001, pp. 41–46.